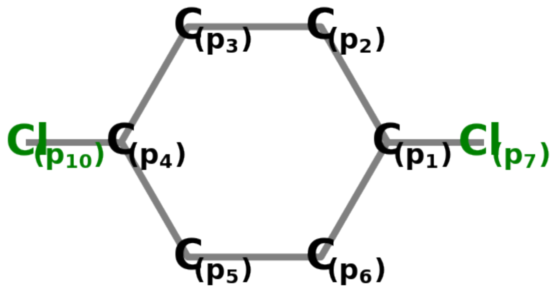
Dichlorobenzene



Point Group: **D2h**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 8E \oplus -4C2(y) \oplus -8\sigma(xy) \oplus 4\sigma(yz)$$

$$\Gamma_{irred} = 1B2g \oplus 3B3g \oplus 1Au \oplus 3B1u$$

- SALC of B3g: $\frac{\sqrt{2}(\mathrm{p}_1-\mathrm{p}_4)}{2}$
- SALC of B1u: $\frac{\sqrt{2}(\mathrm{p}_1+\mathrm{p}_4)}{2}$
- SALC of B2g: $\frac{\mathrm{p}_2}{2}+\frac{\mathrm{p}_3}{2}-\frac{\mathrm{p}_5}{2}-\frac{\mathrm{p}_6}{2}$
- SALC of B3g: $\frac{\mathrm{p}_2}{2}-\frac{\mathrm{p}_3}{2}-\frac{\mathrm{p}_5}{2}+\frac{\mathrm{p}_6}{2}$
- SALC of Au: $\frac{\mathrm{p}_2}{2}-\frac{\mathrm{p}_3}{2}+\frac{\mathrm{p}_5}{2}-\frac{\mathrm{p}_6}{2}$
- SALC of B1u: $\frac{\mathrm{p}_2}{2}+\frac{\mathrm{p}_3}{2}+\frac{\mathrm{p}_5}{2}+\frac{\mathrm{p}_6}{2}$
- SALC of B3g: $\frac{\sqrt{2}(\mathrm{p}_7-\mathrm{p}_8)}{2}$
- SALC of B1u: $\frac{\sqrt{2}(\mathrm{p}_7+\mathrm{p}_8)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{B3g} = \left[\begin{bmatrix} \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 0 \\ \sqrt{2}\tilde{\beta} & \tilde{\alpha}-\tilde{\beta} & 0 \\ 0 & 0 & \tilde{\alpha}_{Cl} \end{bmatrix}\right]$$

$$H_{B1u} = \left[\begin{bmatrix} \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 2\tilde{\beta}_{Cl} \\ \sqrt{2}\tilde{\beta} & \tilde{\alpha}+\tilde{\beta} & 0 \\ 2\tilde{\beta}_{Cl} & 0 & \tilde{\alpha}_{Cl} \end{bmatrix}\right]$$

$$H_{B2g} = \left[\left[\tilde{\alpha}+\tilde{\beta}\right]\right]$$

$$H_{Au} = \left[\left[\tilde{\alpha}-\tilde{\beta}\right]\right]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of B1u with energy = $\frac{\left(4\tilde{\alpha}+2\tilde{\alpha}_{Cl}+2\tilde{\beta}-2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4\left(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}\right.}}\right.}{}$

  (

  $$\begin{bmatrix} ?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B1u} \\ \text{2. SALC in B1u} \\ \text{3. SALC in B1u} \end{bmatrix} = \begin{bmatrix} ?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2 \end{bmatrix} * \begin{bmatrix} \\ \frac{\mathrm{p}_2}{2}+ \\ \end{bmatrix}$$

  $$= ?_1Norm^2\left(\frac{\sqrt{2}\mathrm{p}_1}{2}+\frac{\sqrt{2}\mathrm{p}_4}{2}\right)+?_2Norm^2\left(\frac{\mathrm{p}_2}{2}+\frac{\mathrm{p}_3}{2}+\frac{\mathrm{p}_5}{2}+\frac{\mathrm{p}_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}\mathrm{p}_7}{2}+\frac{\sqrt{2}\mathrm{p}_8}{2}\right)$$

)

- orbital of B1u with energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{}}\right.}{}$

  (

  $$\begin{bmatrix}?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B1u}\\ \text{2. SALC in B1u}\\ \text{3. SALC in B1u}\end{bmatrix} = \begin{bmatrix}?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2\end{bmatrix} * \begin{bmatrix}\sqrt{}\\ \frac{p_2}{2}+\\ \sqrt{}\end{bmatrix}$$

  $$=?_1Norm^2\left(\frac{\sqrt{2}p_1}{2}+\frac{\sqrt{2}p_4}{2}\right)+?_2Norm^2\left(\frac{p_2}{2}+\frac{p_3}{2}+\frac{p_5}{2}+\frac{p_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}p_7}{2}+\frac{\sqrt{2}p_8}{2}\right)$$

)

- orbital of B1u with energy = $\frac{\left(1-\sqrt{3}i\right)\left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}\left(1-\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{}}\right.}{}$

  (

  $$\begin{bmatrix}?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B1u}\\ \text{2. SALC in B1u}\\ \text{3. SALC in B1u}\end{bmatrix} = \begin{bmatrix}?_1Norm^2 & ?_2Norm^2 & ?_3Norm^2\end{bmatrix} * \begin{bmatrix}\sqrt{}\\ \frac{p_2}{2}+\\ \sqrt{}\end{bmatrix}$$

  $$=?_1Norm^2\left(\frac{\sqrt{2}p_1}{2}+\frac{\sqrt{2}p_4}{2}\right)+?_2Norm^2\left(\frac{p_2}{2}+\frac{p_3}{2}+\frac{p_5}{2}+\frac{p_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}p_7}{2}+\frac{\sqrt{2}p_8}{2}\right)$$

)

- orbital of B3g with energy = $\tilde{\alpha}+\tilde{\beta}$
  (

  $$\begin{bmatrix}\frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B3g}\\ \text{2. SALC in B3g}\\ \text{3. SALC in B3g}\end{bmatrix} = \begin{bmatrix}\frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{2}p_1}{2}-\frac{\sqrt{2}p_4}{2}\\ \frac{p_2}{2}-\frac{p_3}{2}-\frac{p_5}{2}+\frac{p_6}{2}\\ \frac{\sqrt{2}p_7}{2}-\frac{\sqrt{2}p_8}{2}\end{bmatrix}$$

  $$=\frac{\sqrt{3}\left(2p_1+p_2-p_3-2p_4-p_5+p_6\right)}{6}$$

)

- orbital of B3g with energy = $\tilde{\alpha}_{Cl}$
  (

  $$\begin{bmatrix}0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\text{1. SALC in B3g}\\ \text{2. SALC in B3g}\\ \text{3. SALC in B3g}\end{bmatrix} = \begin{bmatrix}0 & 0 & 1\end{bmatrix} * \begin{bmatrix}\frac{\sqrt{2}p_1}{2}-\frac{\sqrt{2}p_4}{2}\\ \frac{p_2}{2}-\frac{p_3}{2}-\frac{p_5}{2}+\frac{p_6}{2}\\ \frac{\sqrt{2}p_7}{2}-\frac{\sqrt{2}p_8}{2}\end{bmatrix}$$

  $$=\frac{\sqrt{2}\left(p_7-p_8\right)}{2}$$

)

- orbital of B3g with energy = $\tilde{\alpha} - 2\tilde{\beta}$
  (

$$\begin{bmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B3g} \\ \text{2. SALC in B3g} \\ \text{3. SALC in B3g} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}\mathsf{p}_1}{2} - \frac{\sqrt{2}\mathsf{p}_4}{2} \\ \frac{\mathsf{p}_2}{2} - \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} + \frac{\mathsf{p}_6}{2} \\ \frac{\sqrt{2}\mathsf{p}_7}{2} - \frac{\sqrt{2}\mathsf{p}_8}{2} \end{bmatrix}$$

$$= \frac{\sqrt{6}\left(-\mathsf{p}_1 + \mathsf{p}_2 - \mathsf{p}_3 + \mathsf{p}_4 - \mathsf{p}_5 + \mathsf{p}_6\right)}{6}$$

)

- orbital of B2g with energy = $\tilde{\alpha} + \tilde{\beta}$
  (

$$\begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B2g} \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \frac{\mathsf{p}_2}{2} + \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2} \end{bmatrix}$$

$$= \frac{\mathsf{p}_2}{2} + \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2}$$

)

- orbital of Au with energy = $\tilde{\alpha} - \tilde{\beta}$
  (

$$\begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in Au} \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \frac{\mathsf{p}_2}{2} - \frac{\mathsf{p}_3}{2} + \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2} \end{bmatrix}$$

$$= \frac{\mathsf{p}_2}{2} - \frac{\mathsf{p}_3}{2} + \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2}$$

)

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 8E \ \oplus\ 4C2(y) \ \oplus\ 8\sigma(xy) \ \oplus\ 4\sigma(yz)$$

$$\Gamma_{irred} = 3Ag \ \oplus\ 1B1g \ \oplus\ 3B2u \ \oplus\ 1B3u$$

- SALC of Ag: $\frac{\sqrt{2}(\mathsf{s}_1+\mathsf{s}_4)}{2}$
- SALC of B2u: $\frac{\sqrt{2}(\mathsf{s}_1-\mathsf{s}_4)}{2}$
- SALC of Ag: $\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}+\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}$
- SALC of B1g: $\frac{\mathsf{s}_2}{2}-\frac{\mathsf{s}_3}{2}+\frac{\mathsf{s}_5}{2}-\frac{\mathsf{s}_6}{2}$
- SALC of B2u: $\frac{\mathsf{s}_2}{2}-\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}$
- SALC of B3u: $\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}-\frac{\mathsf{s}_6}{2}$
- SALC of Ag: $\frac{\sqrt{2}(\mathsf{s}_7+\mathsf{s}_8)}{2}$
- SALC of B2u: $\frac{\sqrt{2}(\mathsf{s}_7-\mathsf{s}_8)}{2}$

Put together these SALCs form the following Hamilton matrices:

$$H_{Ag} = \left[\begin{bmatrix} \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 2beta_{\mathsf{s}_Cl} \\ \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s+\tilde{\beta}_s & 0 \\ 2beta_{\mathsf{s}_Cl} & 0 & alpha_{\mathsf{s}_Cl} \end{bmatrix}\right]$$

$$H_{B2u} = \left[\begin{bmatrix} \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 0 \\ \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s-\tilde{\beta}_s & 0 \\ 0 & 0 & alpha_{\mathsf{s}_Cl} \end{bmatrix}\right]$$

$$H_{B1g} = \left[\left[\tilde{\alpha}_s-\tilde{\beta}_s\right]\right]$$

$$H_{B3u} = \left[\left[\tilde{\alpha}_s+\tilde{\beta}_s\right]\right]$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of Ag with energy = $\frac{\left(4\tilde{\alpha}_s+2alpha_{\mathsf{s}_Cl}+2\tilde{\beta}_s-2^{2/3}\sqrt[3]{-27\tilde{\alpha}_s^2alpha_{\mathsf{s}_Cl}-27\tilde{\alpha}_salpha_{\mathsf{s}_Cl}\tilde{\beta}_s+108\tilde{\alpha}_sbeta_{\mathsf{s}_Cl}^2+54alpha_{\mathsf{s}_Cl}\tilde{\beta}_s^2+108\ldots}\right.}{\ldots}$

  (

  $$\left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right] * \begin{bmatrix} \text{1. SALC in Ag} \\ \text{2. SALC in Ag} \\ \text{3. SALC in Ag} \end{bmatrix} = \left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right] * \left[\begin{matrix} \frac{\sqrt{2}\mathsf{s}\ldots}{2} \\ \frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}\ldots}{2}\ldots \\ \frac{\sqrt{2}\mathsf{s}\ldots}{2} \end{matrix}\right.$$

  $$=?_1Norm^2\left(\frac{\sqrt{2}\mathsf{s}_1}{2}+\frac{\sqrt{2}\mathsf{s}_4}{2}\right)+?_2Norm^2\left(\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}+\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}\mathsf{s}_7}{2}+\frac{\sqrt{2}\mathsf{s}_8}{2}\right)$$

)

- orbital of Ag with energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}_s+4alpha_{\mathsf{s}_Cl}+4\tilde{\beta}_s+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}_s^2alpha_{\mathsf{s}_Cl}-27\tilde{\alpha}_salpha_{\mathsf{s}_Cl}\tilde{\beta}_s+108\tilde{\alpha}_sbeta_{\mathsf{s}_Cl}^2+}\right.}{}$

  (

  $$\left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right]*\begin{bmatrix}\text{1. SALC in Ag}\\ \text{2. SALC in Ag}\\ \text{3. SALC in Ag}\end{bmatrix}=\left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right]*\left[\begin{matrix}\frac{\sqrt{2}\mathsf{s}}{2}\\ \frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}}{2}\\ \frac{\sqrt{2}\mathsf{s}}{2}\end{matrix}\right.$$

  $$=?_1Norm^2\left(\frac{\sqrt{2}\mathsf{s}_1}{2}+\frac{\sqrt{2}\mathsf{s}_4}{2}\right)+?_2Norm^2\left(\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}+\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}\mathsf{s}_7}{2}+\frac{\sqrt{2}\mathsf{s}_8}{2}\right)$$

)

- orbital of Ag with energy = $\frac{\left(1-\sqrt{3}i\right)\left(8\tilde{\alpha}_s+4alpha_{\mathsf{s}_Cl}+4\tilde{\beta}_s+2^{2/3}\left(1-\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}_s^2alpha_{\mathsf{s}_Cl}-27\tilde{\alpha}_salpha_{\mathsf{s}_Cl}\tilde{\beta}_s+108\tilde{\alpha}_sbeta_{\mathsf{s}_Cl}^2+}\right.}{}$

  (

  $$\left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right]*\begin{bmatrix}\text{1. SALC in Ag}\\ \text{2. SALC in Ag}\\ \text{3. SALC in Ag}\end{bmatrix}=\left[?_1Norm^2 \quad ?_2Norm^2 \quad ?_3Norm^2\right]*\left[\begin{matrix}\frac{\sqrt{2}\mathsf{s}}{2}\\ \frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}}{2}\\ \frac{\sqrt{2}\mathsf{s}}{2}\end{matrix}\right.$$

  $$=?_1Norm^2\left(\frac{\sqrt{2}\mathsf{s}_1}{2}+\frac{\sqrt{2}\mathsf{s}_4}{2}\right)+?_2Norm^2\left(\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}+\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}\right)+?_3Norm^2\left(\frac{\sqrt{2}\mathsf{s}_7}{2}+\frac{\sqrt{2}\mathsf{s}_8}{2}\right)$$

)

- orbital of B2u with energy = $\tilde{\alpha}_s+\tilde{\beta}_s$

  (

  $$\begin{bmatrix}\frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix}*\begin{bmatrix}\text{1. SALC in B2u}\\ \text{2. SALC in B2u}\\ \text{3. SALC in B2u}\end{bmatrix}=\begin{bmatrix}\frac{\sqrt{6}}{3} & \frac{\sqrt{3}}{3} & 0\end{bmatrix}*\begin{bmatrix}\frac{\sqrt{2}\mathsf{s}_1}{2}-\frac{\sqrt{2}\mathsf{s}_4}{2}\\ \frac{\mathsf{s}_2}{2}-\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}\\ \frac{\sqrt{2}\mathsf{s}_7}{2}-\frac{\sqrt{2}\mathsf{s}_8}{2}\end{bmatrix}$$

  $$=\frac{\sqrt{3}\left(2\mathsf{s}_1+\mathsf{s}_2-\mathsf{s}_3-2\mathsf{s}_4-\mathsf{s}_5+\mathsf{s}_6\right)}{6}$$

)

- orbital of B2u with energy = $alpha_{\mathsf{s}_Cl}$

  (

  $$\begin{bmatrix}0 & 0 & 1\end{bmatrix}*\begin{bmatrix}\text{1. SALC in B2u}\\ \text{2. SALC in B2u}\\ \text{3. SALC in B2u}\end{bmatrix}=\begin{bmatrix}0 & 0 & 1\end{bmatrix}*\begin{bmatrix}\frac{\sqrt{2}\mathsf{s}_1}{2}-\frac{\sqrt{2}\mathsf{s}_4}{2}\\ \frac{\mathsf{s}_2}{2}-\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}+\frac{\mathsf{s}_6}{2}\\ \frac{\sqrt{2}\mathsf{s}_7}{2}-\frac{\sqrt{2}\mathsf{s}_8}{2}\end{bmatrix}$$

  $$=\frac{\sqrt{2}\left(\mathsf{s}_7-\mathsf{s}_8\right)}{2}$$

)

- orbital of B2u with energy = $\tilde{\alpha}_s - 2\tilde{\beta}_s$
  (
  $$\begin{bmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B2u} \\ \text{2. SALC in B2u} \\ \text{3. SALC in B2u} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{3}}{3} & \frac{\sqrt{6}}{3} & 0 \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}\mathsf{s}_1}{2} - \frac{\sqrt{2}\mathsf{s}_4}{2} \\ \frac{\mathsf{s}_2}{2} - \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} + \frac{\mathsf{s}_6}{2} \\ \frac{\sqrt{2}\mathsf{s}_7}{2} - \frac{\sqrt{2}\mathsf{s}_8}{2} \end{bmatrix}$$
  $$= \frac{\sqrt{6}\left(-\mathsf{s}_1 + \mathsf{s}_2 - \mathsf{s}_3 + \mathsf{s}_4 - \mathsf{s}_5 + \mathsf{s}_6\right)}{6}$$
)
- orbital of B3u with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
  (
  $$\begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B3u} \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \frac{\mathsf{s}_2}{2} + \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2} \end{bmatrix}$$
  $$= \frac{\mathsf{s}_2}{2} + \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2}$$
)
- orbital of B1g with energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
  (
  $$\begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B1g} \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} * \begin{bmatrix} \frac{\mathsf{s}_2}{2} - \frac{\mathsf{s}_3}{2} + \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2} \end{bmatrix}$$
  $$= \frac{\mathsf{s}_2}{2} - \frac{\mathsf{s}_3}{2} + \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2}$$
)

Transitions from p orbitals into s orbitals

State with Occupation [2, 2, 2, 0, 0, 0, 2, 2] (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2}:

- $-[2,2,2,0,0,0,2,2],(0,0,0,0,0,0,0,0) \rightarrow (2,1,2,0,0,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1 - \mathsf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1 - \mathsf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)\left(52\tilde{\alpha}+8\tilde{\alpha}_{Cl}+8\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+}\right.}{}$
- Energy of State after Excitation: $\frac{(1-\sqrt{3}i)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s+2\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+1}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-[2,2,2,0,0,0,2,2],(0,0,0,0,0,0,0,0) \rightarrow (2,2,1,0,0,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2} + \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2} + \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{(1-\sqrt{3}i)\left(52\tilde{\alpha}+8\tilde{\alpha}_{Cl}+8\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2+}\right.}{}$
- Energy of State after Excitation: $\frac{(1-\sqrt{3}i)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s+2\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+1}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0, 2, 2) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,1,1,1,1,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,0,1,1,1,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1 - \mathsf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1 - \mathsf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2-}\right.}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{C}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,1,1,1,1,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,0,1,1,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2} + \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2} + \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2-}\right.}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{C}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4}:

- $-(2,1,1,1,1,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,1,0,1,0,2,2),(0,0,0,1,0,0,0,0)$

- ψ_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}\mathsf{p}_1}{2} + \frac{\sqrt{2}\mathsf{p}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{C}}\right.}{}$

- η_4 : linear combination = $3?_1 Norm^2 \left(\frac{\sqrt{2}\mathsf{s}_1}{2} + \frac{\sqrt{2}\mathsf{s}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}_s+4alpha_{\mathsf{s}_{Cl}}+4\tilde{\beta}_s+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-2}\right.}{}$

- Δ Energy: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_{Cl}}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_{Cl}}}{3}-\frac{\tilde{\beta}}{3}+\frac{\tilde{\beta}_s}{3}+\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\frac{\sqrt{-4\left(-3\tilde{\alpha}^2-\right.}}{}}}$

$$\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2alpha_{\mathsf{s}_{Cl}}}{2}-\frac{27\tilde{\alpha}_s alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s beta_{\mathsf{s}_{Cl}}^2+27alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s beta_{\mathsf{s}_{Cl}}^2+\frac{\sqrt{-4\left(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s alpha_{\mathsf{s}_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s+\right.}}{}}}$$

$$\frac{}{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\frac{\sqrt{-4\left(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+\left(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta}\right)^2\right)^3+\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right.}}{}}}$$

$$\frac{}{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2alpha_{\mathsf{s}_{Cl}}}{2}-\frac{27\tilde{\alpha}_s alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s beta_{\mathsf{s}_{Cl}}^2+27alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s^2+54\tilde{\beta}_s beta_{\mathsf{s}_{Cl}}^2+\frac{\sqrt{-4\left(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s alpha_{\mathsf{s}_{Cl}}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_{Cl}}\tilde{\beta}_s+6\right.}}{}}}$$

- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2-}\right.}{}$

- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\sqrt[3]{-27\tilde{\alpha}_s^2 a}}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$

- dipole transition moment: $?_1^2 Norm^4\tilde{\delta}+?_2^2 Norm^4\tilde{\delta}+?_3^2 Norm^4\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5}:

- $-(2,1,1,1,1,0,2,2),(0,0,0,0,0,0,0,0) \to (2,1,1,1,0,0,2,2),(0,0,0,0,1,0,0,0)$

- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$

- η_5 : linear combination = 0, energy = $alpha_{\mathsf{s}_{Cl}}$

- Δ Energy: $-\tilde{\alpha}_{Cl} + alpha_{\mathsf{s}_{Cl}}$

- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(88\tilde{\alpha}+32\tilde{\alpha}_{Cl}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2-}\right.}{}$

- Energy of State after Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(88\tilde{\alpha}+20\tilde{\alpha}_{Cl}+12alpha_{\mathsf{s}_{Cl}}-4\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2-}\right.}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,1,2,0,1,0,2,2),(0,0,0,0,0,0,0,0)\rightarrow(2,0,2,0,1,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1-\mathsf{p}_4)$, energy = $\tilde{\alpha}+\tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1-\mathsf{s}_4)$, energy = $\tilde{\alpha}_s+\tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha}+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s$
- Energy of State before Excitation: $\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right.$
- Energy of State after Excitation: $\left(1-\sqrt{3}i\right)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+}\right.$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,1,2,0,1,0,2,2),(0,0,0,0,0,0,0,0)\rightarrow(2,1,1,0,1,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2}+\frac{\mathsf{p}_3}{2}-\frac{\mathsf{p}_5}{2}-\frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha}+\tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}-\frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s+\tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha}+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s$
- Energy of State before Excitation: $\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right.$
- Energy of State after Excitation: $\left(1-\sqrt{3}i\right)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+}\right.$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5}:

- $-(2,1,2,0,1,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,2,0,0,0,2,2),(0,0,0,0,1,0,0,0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$
- η_5 : linear combination = 0, energy = $alpha_{\mathsf{s}_{Cl}}$
- Δ Energy: $-\tilde{\alpha}_{Cl} + alpha_{\mathsf{s}_{Cl}}$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2 \cdots}\right.}{}$
- Energy of State after Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6alpha_{\mathsf{s}_{Cl}}+2\tilde{\beta}-2\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+ \cdots}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,1,2,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,0,2,1,0,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1 - \mathsf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1 - \mathsf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2 \cdots}\right.}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{C} \cdots\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,1,2,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,1,1,0,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2} + \frac{\mathsf{p}_3}{2} - \frac{\mathsf{p}_5}{2} - \frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2} + \frac{\mathsf{s}_3}{2} - \frac{\mathsf{s}_5}{2} - \frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\Big(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\Big(-27\tilde{\alpha}^2\tilde{\alpha}_{C}}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{C}l}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4}:

- $-(2,1,2,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,2,0,0,0,2,2),(0,0,0,1,0,0,0,0)$
- ψ_4 : linear combination = $3?_1 Norm^2\left(\frac{\sqrt{2}\mathsf{p}_1}{2} + \frac{\sqrt{2}\mathsf{p}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\Big(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{C}}}{}$
- η_4 : linear combination = $3?_1 Norm^2\left(\frac{\sqrt{2}\mathsf{s}_1}{2} + \frac{\sqrt{2}\mathsf{s}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\Big(8\tilde{\alpha}_s+4alpha_{\mathsf{s}_{C}l}+4\tilde{\beta}_s+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-2}}{}$
- Δ Energy: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_{Cl}}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_{C}l}}{3}-\frac{\tilde{\beta}}{3}+\frac{\tilde{\beta}_s}{3}+\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\frac{\sqrt{-4\Big(-3\tilde{\alpha}^2-6}}{}}}$

$$\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2 alpha_{\mathsf{s}_{C}l}}{2}-\frac{27\tilde{\alpha}_s alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s beta_{\mathsf{s}_{C}l}^2+27alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s^2+54\tilde{\beta}_s beta_{\mathsf{s}_{C}l}^2+\frac{\sqrt{-4\Big(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s alpha_{\mathsf{s}_{C}l}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s+}}{}}}$$

$$\frac{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\frac{\sqrt{-4\left(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+\left(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta}\right)^2\right)^3+\Big(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}}{}}}{}$$

$$\frac{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2 alpha_{\mathsf{s}_{C}l}}{2}-\frac{27\tilde{\alpha}_s alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s}{2}+54\tilde{\alpha}_s beta_{\mathsf{s}_{C}l}^2+27alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s^2+54\tilde{\beta}_s beta_{\mathsf{s}_{C}l}^2+\frac{\sqrt{-4\Big(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_s alpha_{\mathsf{s}_{C}l}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_{C}l}\tilde{\beta}_s+6}}{}}}{}$$

- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\Big(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}}{}$

- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\sqrt[3]{-27\tilde{\alpha}_s^2 a}}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $?_1^2 Norm^4\tilde{\delta}+?_2^2 Norm^4\tilde{\delta}+?_3^2 Norm^4\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,2,1,0,1,0,2,2),(0,0,0,0,0,0,0,0)\rightarrow(2,1,1,0,1,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1-\mathsf{p}_4)$, energy = $\tilde{\alpha}+\tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1-\mathsf{s}_4)$, energy = $\tilde{\alpha}_s+\tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha}+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right.}{}$

- Energy of State after Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+}\right.}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,2,1,0,1,0,2,2),(0,0,0,0,0,0,0,0)\rightarrow(2,2,0,0,1,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2}+\frac{\mathsf{p}_3}{2}-\frac{\mathsf{p}_5}{2}-\frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha}+\tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}-\frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s+\tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha}+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}\right.}{}$

- Energy of State after Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(40\tilde{\alpha}+14\tilde{\alpha}_{Cl}+6\tilde{\alpha}_s-4\tilde{\beta}+6\tilde{\beta}_s-2\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+}\right.}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5}:

- $-(2,2,1,0,1,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,2,1,0,0,0,2,2),(0,0,0,0,1,0,0,0)$
- ψ_5 : linear combination = 0, energy = $\tilde{\alpha}_{Cl}$
- η_5 : linear combination = 0, energy = $alpha_{\mathsf{s}_{Cl}}$
- Δ Energy: $-\tilde{\alpha}_{Cl} + alpha_{\mathsf{s}_{Cl}}$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\Bigg(46\tilde{\alpha}+14\tilde{\alpha}_{Cl}+2\tilde{\beta}-2\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+108\tilde{\beta}\tilde{\beta}_{Cl}^2}}{}$
- Energy of State after Excitation: $\frac{\left(1-\sqrt{3}i\right)\Bigg(46\tilde{\alpha}+8\tilde{\alpha}_{Cl}+6alpha_{\mathsf{s}_{Cl}}+2\tilde{\beta}-2\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2+}}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$
- dipole transition moment: $\tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0, 2, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2}:

- $-(2,2,1,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,1,1,1,0,0,2,2),(0,1,0,0,0,0,0,0)$
- ψ_2 : linear combination = $\sqrt{3}\,(\mathsf{p}_1 - \mathsf{p}_4)$, energy = $\tilde{\alpha} + \tilde{\beta}$
- η_2 : linear combination = $\sqrt{3}\,(\mathsf{s}_1 - \mathsf{s}_4)$, energy = $\tilde{\alpha}_s + \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\Bigg(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\Bigg(-27\tilde{\alpha}^2\tilde{\alpha}_{C}}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4} - \frac{\tilde{\alpha}_{Cl}}{4} + \frac{3\tilde{\alpha}_s}{4} + \frac{alpha_{\mathsf{s}_{Cl}}}{4}$

- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3}:

- $-(2,2,1,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,2,0,1,0,0,2,2),(0,0,1,0,0,0,0,0)$
- ψ_3 : linear combination = $\frac{\mathsf{p}_2}{2}+\frac{\mathsf{p}_3}{2}-\frac{\mathsf{p}_5}{2}-\frac{\mathsf{p}_6}{2}$, energy = $\tilde{\alpha}+\tilde{\beta}$
- η_3 : linear combination = $\frac{\mathsf{s}_2}{2}+\frac{\mathsf{s}_3}{2}-\frac{\mathsf{s}_5}{2}-\frac{\mathsf{s}_6}{2}$, energy = $\tilde{\alpha}_s+\tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha}+\tilde{\alpha}_s-\tilde{\beta}+\tilde{\beta}_s$
- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}\right.}{}$
- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\left(-27\tilde{\alpha}^2\tilde{\alpha}_{C}\right.}{}$
- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_Cl}}{4}$
- dipole transition moment: $\tilde{\delta}$

\- AND

Transition from ψ_4 into ψ_{s_4}:

- $-(2,2,1,1,0,0,2,2),(0,0,0,0,0,0,0,0) \rightarrow (2,2,1,0,0,0,2,2),(0,0,0,1,0,0,0,0)$
- ψ_4 : linear combination = $3?_1Norm^2\left(\frac{\sqrt{2}\mathsf{p}_1}{2}+\frac{\sqrt{2}\mathsf{p}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}+4\tilde{\alpha}_{Cl}+4\tilde{\beta}+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{C}}\right.}{}$
- η_4 : linear combination = $3?_1Norm^2\left(\frac{\sqrt{2}\mathsf{s}_1}{2}+\frac{\sqrt{2}\mathsf{s}_4}{2}\right)$, energy = $\frac{\left(1+\sqrt{3}i\right)\left(8\tilde{\alpha}_s+4alpha_{\mathsf{s}_Cl}+4\tilde{\beta}_s+2^{2/3}\left(1+\sqrt{3}i\right)\sqrt[3]{-2}\right.}{}$
- Δ Energy: $-\frac{2\tilde{\alpha}}{3}-\frac{\tilde{\alpha}_{Cl}}{3}+\frac{2\tilde{\alpha}_s}{3}+\frac{alpha_{\mathsf{s}_Cl}}{3}-\frac{\tilde{\beta}}{3}+\frac{\tilde{\beta}_s}{3}+\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4\left(-3\tilde{\alpha}^2-\right.}}}$

$$\frac{}{3\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2alpha_{\mathsf{s}_Cl}}{2}-\frac{27\tilde{\alpha}_salpha_{\mathsf{s}_Cl}\tilde{\beta}_s}{2}+54\tilde{\alpha}_sbeta_{\mathsf{s}_Cl}^2+27alpha_{\mathsf{s}_Cl}\tilde{\beta}_s^2+54\tilde{\beta}_sbeta_{\mathsf{s}_Cl}^2+\sqrt{-4\left(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_salpha_{\mathsf{s}_Cl}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_Cl}\tilde{\beta}_s+\right.}}}$$

$$\frac{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}^2\tilde{\alpha}_{Cl}}{2}-\frac{27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}}{2}+54\tilde{\alpha}\tilde{\beta}_{Cl}^2+27\tilde{\alpha}_{Cl}\tilde{\beta}^2+54\tilde{\beta}\tilde{\beta}_{Cl}^2+\sqrt{-4\left(-3\tilde{\alpha}^2-6\tilde{\alpha}\tilde{\alpha}_{Cl}-3\tilde{\alpha}\tilde{\beta}-3\tilde{\alpha}_{Cl}\tilde{\beta}+6\tilde{\beta}^2+12\tilde{\beta}_{Cl}^2+\left(-2\tilde{\alpha}-\tilde{\alpha}_{Cl}-\tilde{\beta}\right)^2\right)^3+\left(-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}\right.}}}{}$$

$$\frac{\left(\frac{1}{2}+\frac{\sqrt{3}i}{2}\right)\sqrt[3]{-\frac{27\tilde{\alpha}_s^2alpha_{\mathsf{s}_Cl}}{2}-\frac{27\tilde{\alpha}_salpha_{\mathsf{s}_Cl}\tilde{\beta}_s}{2}+54\tilde{\alpha}_sbeta_{\mathsf{s}_Cl}^2+27alpha_{\mathsf{s}_Cl}\tilde{\beta}_s^2+54\tilde{\beta}_sbeta_{\mathsf{s}_Cl}^2+\sqrt{-4\left(-3\tilde{\alpha}_s^2-6\tilde{\alpha}_salpha_{\mathsf{s}_Cl}-3\tilde{\alpha}_s\tilde{\beta}_s-3alpha_{\mathsf{s}_Cl}\tilde{\beta}_s+6\right.}}}{}$$

- Energy of State before Excitation: $\frac{\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(100\tilde{\alpha}+20\tilde{\alpha}_{Cl}+8\tilde{\beta}-4\cdot 2^{2/3}\sqrt[3]{-27\tilde{\alpha}^2\tilde{\alpha}_{Cl}-27\tilde{\alpha}\tilde{\alpha}_{Cl}\tilde{\beta}+108\tilde{\alpha}\tilde{\beta}_{Cl}^2+54\tilde{\alpha}_{Cl}\tilde{\beta}^2}\right.}{}$

- Energy of State after Excitation: $\frac{8\sqrt[3]{2}\left(1-\sqrt{3}i\right)\left(1+\sqrt{3}i\right)\left(3\tilde{\alpha}^2+6\tilde{\alpha}\tilde{\alpha}_{Cl}+3\tilde{\alpha}\tilde{\beta}+3\tilde{\alpha}_{Cl}\tilde{\beta}-6\tilde{\beta}^2-12\tilde{\beta}_{Cl}^2-\left(2\tilde{\alpha}+\tilde{\alpha}_{Cl}+\tilde{\beta}\right)^2\right)\sqrt[3]{-27\tilde{\alpha}_s^2 a}}{}$

- Energy between Average Energies of ψ-/η-orbitals: $-\frac{3\tilde{\alpha}}{4}-\frac{\tilde{\alpha}_{Cl}}{4}+\frac{3\tilde{\alpha}_s}{4}+\frac{alpha_{\mathsf{s}_C l}}{4}$
- dipole transition moment: $?_1^2 Norm^4\tilde{\delta}+?_2^2 Norm^4\tilde{\delta}+?_3^2 Norm^4\tilde{\delta}$

- AND

Dispersion energies following from the transitions

1. State: A = [2, 2, 2, 0, 0, 0, 2, 2] ↔ B = (2, 1, 1, 1, 1, 0, 2, 2)

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{32\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6\left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3}\right.}$$

2. State: A = (2, 1, 1, 1, 1, 0, 2, 2) ↔ B = [2, 2, 2, 0, 0, 0, 2, 2]

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{32\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6\left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3}\right.}$$

3. State: A = (2, 1, 2, 0, 1, 0, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4\left(-12\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right) - 9\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - alph\right.\right.}{r^6\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)\left(\tilde{\alpha} +\right.}$$

4. State: A = (2, 1, 2, 0, 1, 0, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6\left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_C}{3}\right.}$$

5. State: A = (2, 1, 2, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4\left(-12\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right) - 9\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - alph\right.\right.}{r^6\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)\left(\tilde{\alpha} +\right.}$$

6. State: A = (2, 1, 2, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6\left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_C}{3}\right.}$$

7. State: A = (2, 1, 2, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{\quad}{r^6\left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3} \cdots\right.}$$

8. State: A = (2, 1, 2, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{\quad}{r^6\left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{\quad}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)\sqrt[3]{- \cdots}} \right.}$$

9. State: A = (2, 1, 2, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{\quad}{r^6\left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3} \cdots\right.}$$

10. State: A = (2, 1, 2, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{\quad}{r^6\left(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{\quad}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)\sqrt[3]{- \cdots}} \right.}$$

11. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4\left(-12\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right) - 9\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - alph \cdots\right.\right.}{r^6\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)\left(\tilde{\alpha} + \cdots\right.}$$

12. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{\quad}{r^6\left(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3} \cdots\right.}$$

13. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{2\tilde{\delta}^4 \left(-12\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right) - 9\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)(\tilde{\alpha} + \tilde{\alpha}_{Cl} - \tilde{\alpha}_s - alph}{r^6\left(\tilde{\alpha}_{Cl} - alpha_{\mathsf{s}_{Cl}}\right)\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)(\tilde{\alpha} +}$$

14. State: A = (2, 2, 1, 0, 1, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3}}$$

15. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3}}$$

16. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 1, 2, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)\sqrt[3]{-}}}$$

17. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2, 2)

$$E_{dispersion} = \frac{12\tilde{\delta}^4}{r^6\left(-\tilde{\alpha} - \tilde{\alpha}_{Cl} + \tilde{\alpha}_s + alpha_{\mathsf{s}_{Cl}} - \tilde{\beta} + \tilde{\beta}_s\right)} + \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6(-\frac{2\tilde{\alpha}}{3} - \frac{4\tilde{\alpha}_{C}}{3}}$$

18. State: A = (2, 2, 1, 1, 0, 0, 2, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2, 2)

$$E_{dispersion} = \frac{36\tilde{\delta}^4}{r^6\left(-2\tilde{\alpha} + 2\tilde{\alpha}_s - 2\tilde{\beta} + 2\tilde{\beta}_s\right)} + \frac{}{r^6(-\frac{5\tilde{\alpha}}{3} - \frac{\tilde{\alpha}_{Cl}}{3} + \frac{5\tilde{\alpha}_s}{3} + \frac{alpha_{\mathsf{s}_{Cl}}}{3} - \frac{4\tilde{\beta}}{3} + \frac{4\tilde{\beta}_s}{3} + \frac{}{3\left(-\frac{1}{2} - \frac{\sqrt{3}i}{2}\right)\sqrt[3]{-}}}$$

Ground-State:
A =[2, 2, 2, 0, 0, 0, 2, 2]↔ B =[2, 2, 2, 0, 0, 0, 2, 2]

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6\left(\tilde{\alpha} - \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s\right)}$$